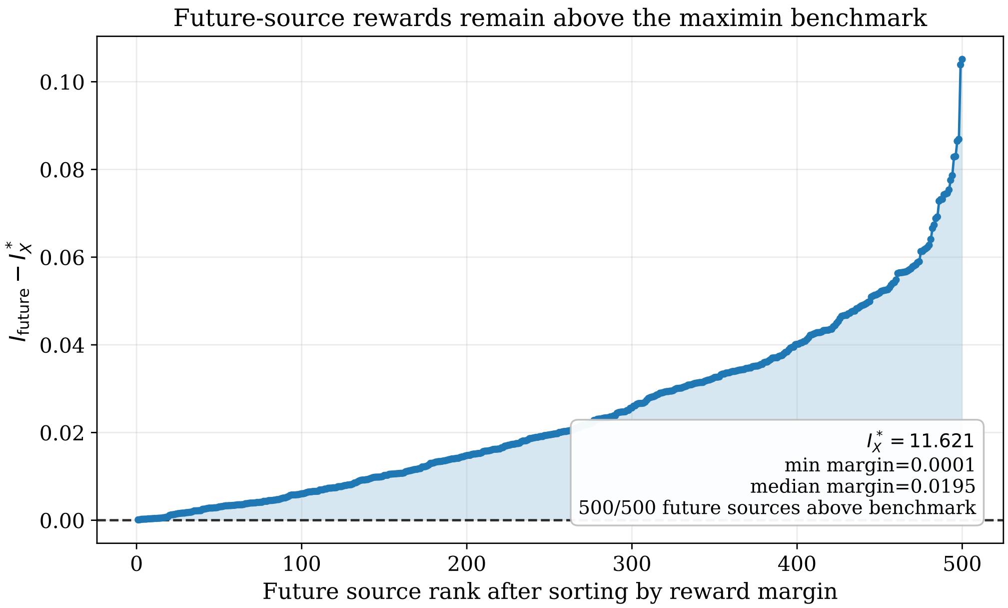


Figure 3: Future-source conservativeness in Simulation 6. Each point is the reward margin $I_{\text{future}} - I_X^*$ for one future source after sorting the 500 margins. The dashed horizontal line marks zero. All future sources have nonnegative margins, supporting the conservative interpretation of the maximin benchmark for mechanisms in the convex hull of the training sources.

6 Real-data analysis: stable erythroid GEX–ADT associations in BMMC CITE-seq

We analysed the human bone marrow mononuclear cell (BMMC) CITE-seq data from the NeurIPS 2021 Multimodal Single-Cell Data Integration challenge (Lance et al., 2022; Luecken et al., 2021; Gene Expression Omnibus, 2022). The original benchmark was designed for multimodal single-cell prediction and integration. We use it for a different inferential target: among candidate RNA features, which genes provide a stable explanation of erythroid surface-protein abundance across related cell states? CITE-seq is well suited to this question because it measures gene expression and antibody-derived tags (ADTs) in the same cell (Stoeckius et al., 2017). Bone marrow is also a natural multi-source system: erythroid cells appear along a developmental trajectory, so a gene that is predictive only in one stage is less biologically transportable than a gene whose contribution persists across several erythroid stages.

We used the preprocessed file `Redcell_selected_cell_types_by_batch_hvg1000.npz`. The analysis retained 11,948 cells, 1000 highly variable genes (HVGs) and 134 ADTs. HVG filtering was performed within the selected red-cell-related subset using the Seurat-v3 procedure implemented in `scanpy`, so that the candidate set focused on genes with substantial variation in the relevant erythroid module (Stuart et al., 2019; Wolf et al., 2018). The source domains were Proerythroblast, Erythroblast and Normoblast; Reticulocyte and

Table 5: OOD-cell-type R^2 of top-20 selected genes for the four primary erythroid ADTs.

ADT	OOD cell type	MIMAL	Fisher	Pooling	Max-p
CD105	MK/E prog	0.624	0.654	**0.673**	0.651
CD105	Reticulocyte	**0.238**	0.134	0.167	0.145
CD36	MK/E prog	0.660	0.506	0.565	**0.677**
CD36	Reticulocyte	**0.470**	0.332	0.180	0.366
CD49d	MK/E prog	**0.737**	0.716	0.715	0.676
CD49d	Reticulocyte	**0.558**	0.453	0.415	0.471
CD71	MK/E prog	**0.574**	0.536	0.500	0.456
CD71	Reticulocyte	**0.344**	0.315	0.316	0.230

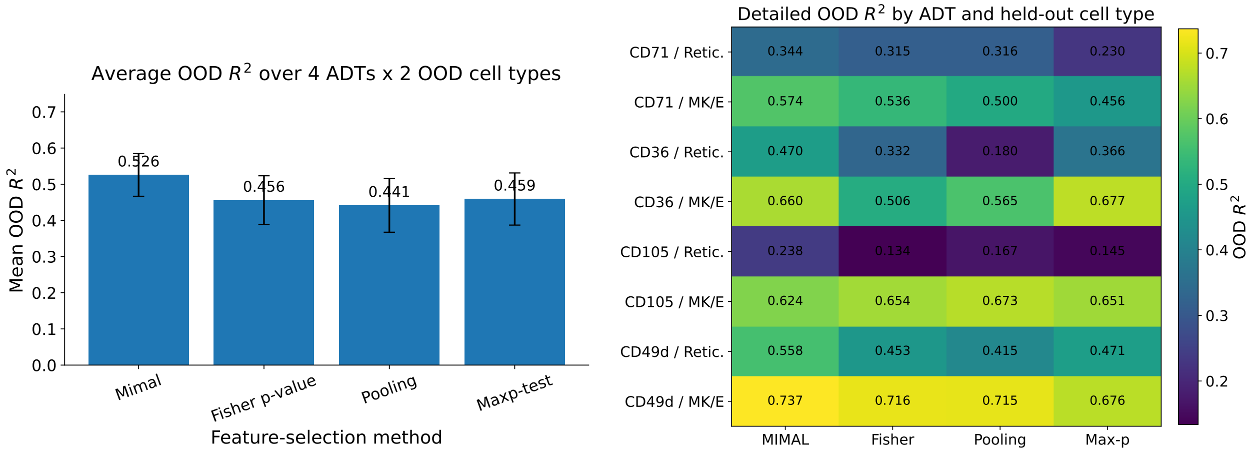
The best method in each ADT–OOD row is shown in bold. A row compares methods under the same ADT, the same held-out cell type and the same top-20 feature budget. Reticulocyte is the terminal erythroid OOD state; MK/E prog is an upstream progenitor-adjacent OOD state near the megakaryocyte–erythroid branch.

MK/E prog were held out from feature selection and used only for out-of-distribution (OOD) evaluation. The four primary ADTs were CD71, CD36, CD105 and CD49d. These proteins represent complementary erythroid biology: CD71 is the transferrin receptor and reflects iron uptake; CD36 is an early erythroid and metabolic marker; CD105 is associated with early erythroid/progenitor states; and CD49d/ITGA4 is an adhesion-related integrin relevant to bone-marrow niche interaction (Mori et al., 2015; Machherndl-Spandl et al., 2013; Yan et al., 2021; Ulyanova et al., 2014).

For each ADT, MIMAL was compared with pooling, Fisher p-value aggregation and max-p testing under the same 1000-gene candidate set and the same source/OOD split. The implementation used the squared-error explicit solution Algorithm B3, where the shared signal is the adversarially aggregated residualized source effect rather than the raw source conditional mean. Appendix C gives the exact source sizes, the top genes and additional diagnostic figures.

Table 5 reports the post-selection OOD R^2 for each ADT and held-out cell type. Each number is computed after selecting the top 20 genes by a given method in the three source cell types and then fitting those genes to the same ADT in the held-out cell type. Thus, the table does not measure a prospective clinical prediction model; it measures whether the selected genes remain explanatory in a biologically related cell state that did not participate in feature selection. Figure 4 summarizes the same comparison: panel (a) averages over the eight ADT–OOD combinations, whereas panel (b) shows the full ADT-by-OOD matrix.

Across the eight comparisons, the average OOD R^2 was 0.526 for MIMAL, compared with 0.456 for Fisher aggregation, 0.441 for pooling and 0.459 for max-p testing. More importantly, MIMAL won all four comparisons in Reticulocyte. This is the cleanest biological validation in the analysis, because Reticulocyte is a terminal erythroid state not used during



(a) Average over all eight ADT–OOD comparisons.

(b) Detailed ADT-by-OOD comparison.

Figure 4: Post-selection OOD R^2 comparison. Panel (a) shows that MIMAL has the largest average OOD R^2 across the four ADTs and two held-out cell types. Panel (b) identifies where the gains arise: MIMAL is best for all four ADTs in Reticulocyte and is also best for CD71 and CD49d in MK/E prog. The two non-MIMAL wins occur in MK/E prog, where progenitor-branch-specific variation is expected to be stronger.

selection. The result suggests that MIMAL is not simply choosing genes that fit one nucleated source stage; it is selecting genes whose residualized contribution continues to explain protein abundance after erythroid maturation has progressed. CD49d provides the strongest example, with $R^2 = 0.558$ in Reticulocyte and $R^2 = 0.737$ in MK/E prog, consistent with a stable adhesion and niche-interaction programme. CD36 also shows a large Reticulocyte advantage for MIMAL, whereas pooling performs poorly, indicating that average-source ranking can select genes that fit the pooled source composition but transfer weakly to the terminal erythroid target. CD105 is the main exception in MK/E prog, where pooling is best; this is biologically plausible because CD105 is more progenitor-biased and MK/E prog is close to the upstream megakaryocyte–erythroid branch.

To interpret the selected genes, we performed Gene Ontology (GO) over-representation analysis with the GO/PANTHER tool (The Gene Ontology Consortium, 2021; Mi et al., 2019). For each query, the reference set was the same 1000-HVG list used by the feature-selection methods, rather than all human genes. This background is important: all methods were constrained to choose genes from the HVG candidate set, so enrichment should be evaluated relative to that same candidate universe. Figure 5 displays representative significant GO terms for the MIMAL top-20 genes, and Table 6 gives the corresponding counts, expected counts, fold enrichments and FDR values.

The enrichment results are consistent with the OOD prediction results. CD49d, the ADT with the strongest OOD R^2, also has the clearest functional annotation. Its MIMAL-selected genes are enriched for oxidative phosphorylation, electron transport chain, respiratory-chain

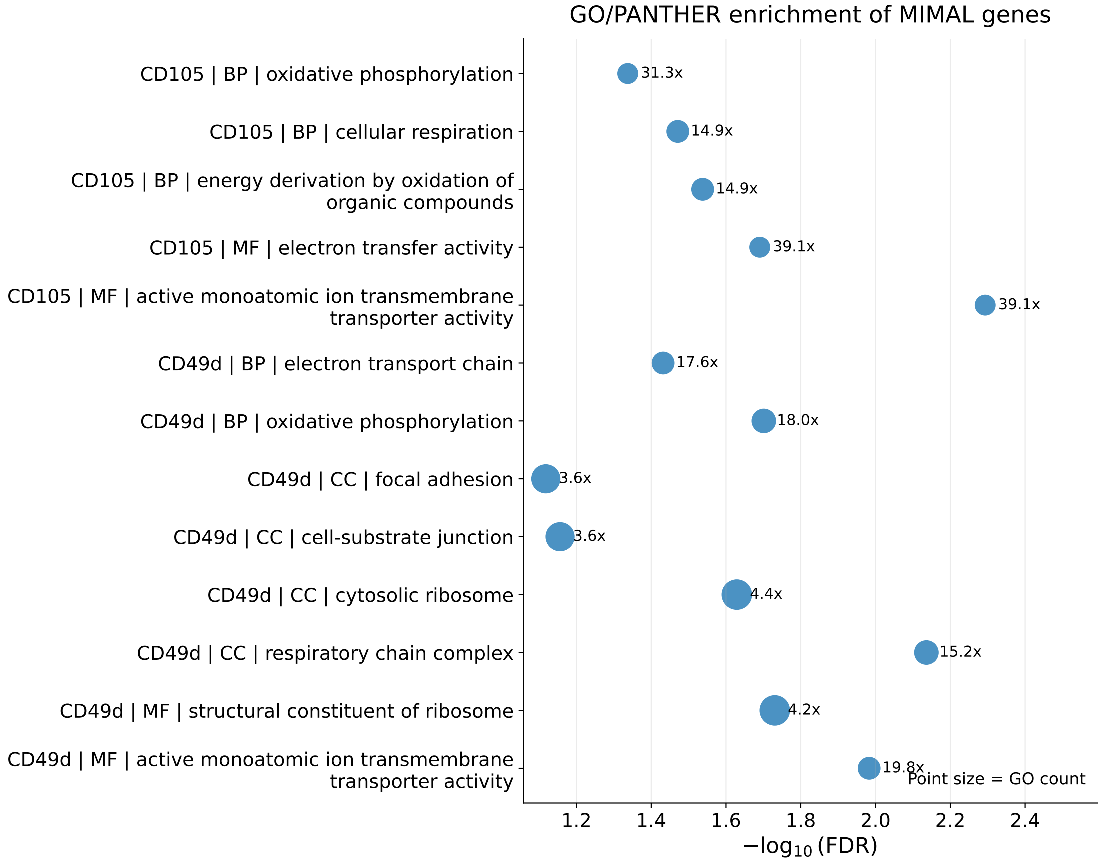


Figure 5: GO/PANTHER over-representation analysis for MIMAL-selected genes. Each point is a significant over-represented GO term from a protein-specific MIMAL top-20 list. The horizontal axis is $-\log_{10}(\mathrm{FDR})$, the point size is proportional to the number of selected genes annotated to the term, and the text next to each point gives the fold enrichment. The dominant signals are respiratory-chain/oxidative-phosphorylation terms and ribosomal/translation terms, with adhesion-related terms appearing for CD49d.

complex, cytosolic ribosome, structural constituent of ribosome, focal adhesion and cell-substrate junction. This combination is biologically interpretable: CD49d is an integrin-related surface marker, so an explanatory RNA signature involving cell-substrate adhesion is directly relevant, while the ribosomal and respiratory-chain terms reflect the large translational and metabolic reprogramming accompanying erythroid maturation. CD105 also shows significant enrichment for energy derivation, cellular respiration, oxidative phosphorylation and transporter activity, consistent with an early erythroid/progenitor marker whose abundance is linked to metabolic state. In contrast, CD71 and CD36 did not have protein-specific top-20 GO terms passing FDR < 0.10 under the HVG background. We interpret this conservatively: the absence of significant enrichment for these two ADTs does not negate their OOD predictiveness; rather, it reflects the limited size of each top-20 query list and the fact that the analysis tests only annotations available within the HVG-restricted candidate universe.

Table 6: Representative GO/PANTHER enrichment terms for MIMAL-selected genes.

ADT	Ontology	GO term	Count	Expected	Fold	FDR
CD105	BP	energy derivation by oxidation of organic compounds (GO:0015980)	4	0.27	14.89	0.029
CD105	BP	cellular respiration (GO:0045333)	4	0.27	14.89	0.0338
CD105	BP	oxidative phosphorylation (GO:0006119)	3	0.10	31.26	0.046
CD105	MF	active monoatomic ion transmembrane transporter activity (GO:0022853)	3	0.08	39.08	0.00509
CD105	MF	electron transfer activity (GO:0009055)	3	0.08	39.08	0.0204
CD49d	BP	oxidative phosphorylation (GO:0006119)	5	0.28	18.00	0.0199
CD49d	BP	electron transport chain (GO:0022900)	4	0.23	17.60	0.037
CD49d	CC	respiratory chain complex (GO:0098803)	5	0.33	15.23	0.00731
CD49d	CC	cytosolic ribosome (GO:0022626)	9	2.05	4.40	0.0235
CD49d	CC	cell-substrate junction (GO:0030055)	8	2.22	3.60	0.0698
CD49d	CC	focal adhesion (GO:0005925)	8	2.22	3.60	0.0762
CD49d	MF	active monoatomic ion transmembrane transporter activity (GO:0022853)	4	0.20	19.80	0.0104
CD49d	MF	structural constituent of ribosome (GO:0003735)	9	2.12	4.24	0.0186

BP, biological process; CC, cellular component; MF, molecular function. "Count" is the number of MIMAL-selected top-20 genes annotated to the term, and "Expected" is the corresponding count expected under the 1000-HVG reference background. Only representative over-represented terms with FDR below 0.10 are shown.

Together, the two analyses support the intended scientific use of MIMAL. The R^2 comparison shows that MIMAL-selected genes transfer better to held-out erythroid states, especially Reticulocyte. The GO/PANTHER analysis shows that the transferred signal is not arbitrary high-dimensional prediction noise, but is enriched for biologically plausible programmes involving mitochondrial respiration, translation and adhesion. Thus, in this BMMC application, MIMAL prioritizes RNA features whose contribution to ADT explainability is stable across erythroid source domains, rather than features that are strong only in one source or only under a pooled average criterion.

7 Conclusion and Discussion

In this paper, we introduced MIMAL, a novel framework for assessing stable variable importance across heterogeneous data sources via adversarial learning. By optimizing the worst-case predictive reward among the sources, MIMAL quantifies variable importance in a manner that emphasizes generalizability and stability. The framework incorporates advanced ML techniques and ensures asymptotic unbiasedness and normality of $\widehat{I_X}$, even under complex data and model settings. Extensive simulations and real-world application highlight its utility in diverse scenarios, showcasing its capacity to address challenges in multi-source variable importance analysis with interpretability and reliability.

We now discuss several future directions to improve and generalize our work. Firstly, it is potentially interesting to extend our framework to a transfer learning setting with covariate shift. In this scenario, we are interested in quantifying the predictive importance of X on some target population $\mathbb{Q}_{X,Z} \neq \mathbb{P}^{(m)}_{X,Z}$'s. Meanwhile, on the target, we only have samples of

Zhang, Z., Zhan, W., Chen, Y., Du, S. S., and Lee, J. D. (2024). Optimal multi-distribution learning. In *The Thirty Seventh Annual Conference on Learning Theory*, pages 5220–5223. PMLR.

Zhou, D.-X. (2008). Derivative reproducing properties for kernel methods in learning theory. *Journal of Computational and Applied Mathematics*, 220(1):456–463.

For squared-error reward under the common covariate law, the reward of the learned maximin coefficient on a future source is

$$I_{\text{future}} = 12\{2\bar{\beta}^\top \beta_{\text{future}} - \bar{\beta}^\top \bar{\beta}\}.$$

The maximin projection geometry implies $I_{\text{future}} \geq I_X^*$ for future coefficients in the convex hull; Simulation 6 checks this numerically over 500 future sources.

Unequal source-size design, Simulation 7. Simulation 7 uses the same five source mechanisms as Simulation 6 but source sample sizes

$$(n_1, n_2, n_3, n_4, n_5) = (300, 1000, 3000, 800, 1800).$$

Covariate shift is generated through the same softmax source-assignment mechanism, with production grid $\alpha \in \{0.05, 0.10, 0.20\}$. The estimator is DA-MIMAL with the augmented reward and the variance estimator that contains both the target-sample contribution and the source-specific residual contributions. This design is intended to check whether the standard error correctly reflects highly imbalanced source information.

C Additional details for the BMMC CITE-seq application

C.1 Source/OOD design and biological rationale

Table B1 gives the exact cell types used in the erythroid application. The three source domains are all nucleated erythroid stages, but they are not identical: Proerythroblast is early, Erythroblast is intermediate and Normoblast is close to enucleation. This makes the maximin question meaningful. A gene must contribute across several related erythroid environments to receive a high MIMAL score. The two OOD cell types test two different forms of transfer: Reticulocyte tests terminal erythroid transfer, whereas MK/E prog tests transfer toward the upstream progenitor-adjacent branch.

C.2 Implementation details

For each ADT and candidate gene, the explicit squared-error MIMAL procedure was run under the residualized source aggregation described in Section **??**. In this implementation, the gene under testing was treated as the exposure feature, and the remaining high-dimensional GEX information was summarized as a background adjustment. This implements the intended LOCO-style interpretation: the score asks whether the gene adds stable predictive information for the ADT beyond a broad expression-background model. Pooling, Fisher p-value aggregation and max-p testing were run on the same candidate set and source split.

Table B1: Cell-type roles in the erythroid BMMC application.

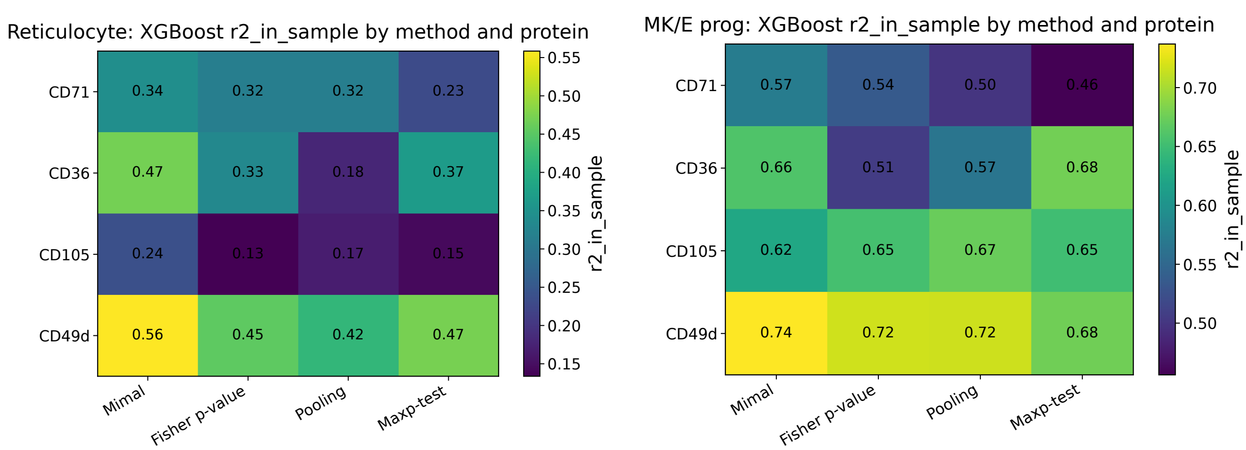
Role	Cell type	Cells	Interpretation
Source	Proerythroblast	1512	Early erythroid precursor; onset of the erythroid transcriptional programme.
Source	Erythroblast	4039	Intermediate erythroid stage with strong haemoglobin, iron-use and biosynthetic activity.
Source	Normoblast	1435	Mature nucleated erythroid precursor close to enucleation.
OOD	Reticulocyte	4272	Terminal erythroid state after enucleation and before mature red blood cell.
OOD	MK/E prog	690	Megakaryocyte–erythroid progenitor near lineage branching.

The OOD cell types were not used for feature selection. Their role is to assess whether a selected RNA signature generalizes beyond the three erythroid source states.

For downstream comparison, the top 20 genes from each method were evaluated in each OOD cell type using the same ADT outcome.

C.3 Additional OOD R^2 figures

Figure 9 shows the OOD R^2 matrices separately for Reticulocyte and MK/E prog using the large-font figures from the updated red-cell analysis archive. These heatmaps are the detailed version of the main-text summary. The Reticulocyte panel is especially important because it tests transfer to a later erythroid stage absent from selection; MIMAL is highest for all four ADTs in that panel.

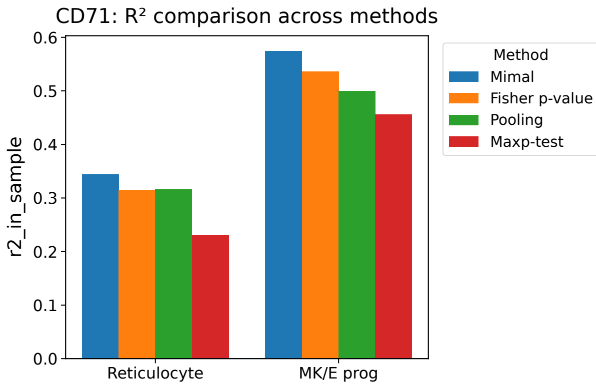


(a) Reticulocyte OOD.

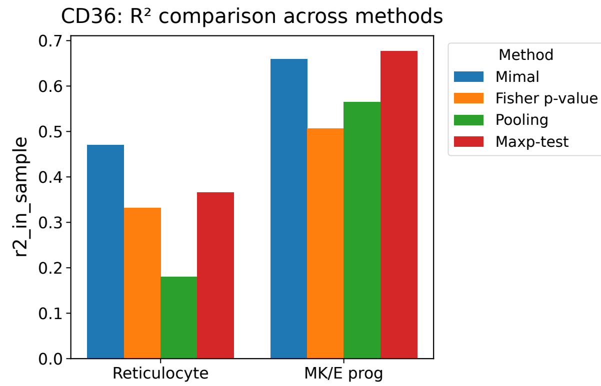
(b) MK/E prog OOD.

Figure 9: Detailed OOD R^2 heatmaps by held-out cell type. Each entry is the R^2 obtained by fitting the top-20 genes selected by one method to the ADT in the held-out cell type. Reticulocyte evaluates terminal erythroid transfer, whereas MK/E prog evaluates transfer to a progenitor-adjacent branch.

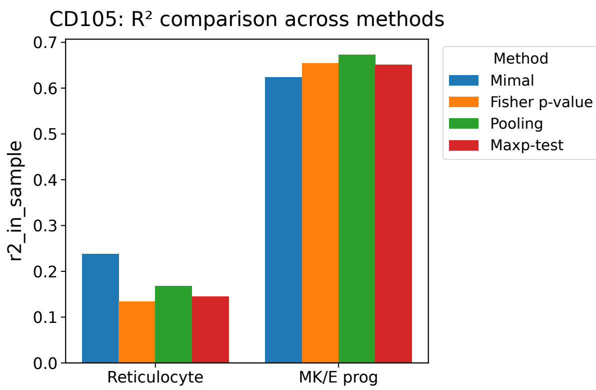
Figure 10 gives per-ADT bar plots. These plots are useful for diagnosing whether MIMAL's overall advantage is driven by one protein or is repeated across proteins. They show that the advantage is most pronounced for CD49d and CD36 in Reticulocyte and is still present for CD71 and CD105 in that same terminal OOD state.



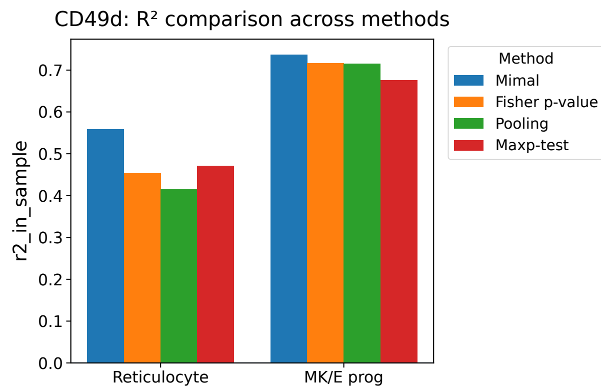
(a) CD71.



(b) CD36.



(c) CD105.



(d) CD49d.

Figure 10: Per-protein OOD R^2 bar plots. Each panel fixes one ADT and compares methods in the two held-out cell types. The plots show that MIMAL's advantage is not only an average effect; it is repeated across all four proteins in Reticulocyte.

C.4 Selected genes and method overlap

Table B2 lists the first ten MIMAL-ranked genes for each primary ADT. The lists contain ribosomal genes, mitochondrial genes and cytoskeleton/adhesion-related genes such as *ACTB*. This is consistent with the GO enrichment in the main text. The selected genes should not be interpreted as a complete catalogue of erythroid causal regulators: the candidate set is restricted to 1000 HVGs and each MIMAL score is adjusted for a high-dimensional background expression programme.

Table B2: First ten MIMAL-ranked genes for each primary ADT.

ADT	First ten MIMAL-ranked genes
CD71	MTRNR2L8, RPS10, ACTB, MT-CO3, RPS8, RPS4X, RPL13A, CPVL, SYNGR1, SNHG29
CD36	MTRNR2L8, ACTB, RPS10, HIST1H2AL, LYZ, HMGN1, AC024267.1, NME2, RPS26, RPL10A
CD105	MTRNR2L8, MT-CO2, ACTB, FCER1A, MT-CO3, TMCC2, LYZ, MT-CYB, PKM, TRDC
CD49d	MTRNR2L8, RPS12, MT-CO3, RPL36, AC024267.1, RPS10, MT-CYB, MT-CO2, HIST1H4H, RPL37A

These genes are selected within the HVG-restricted candidate universe. They are useful for interpreting the stable predictive signature but should not be read as an exhaustive erythroid marker list.